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InclusiFinance Loan Default Risk Analysis

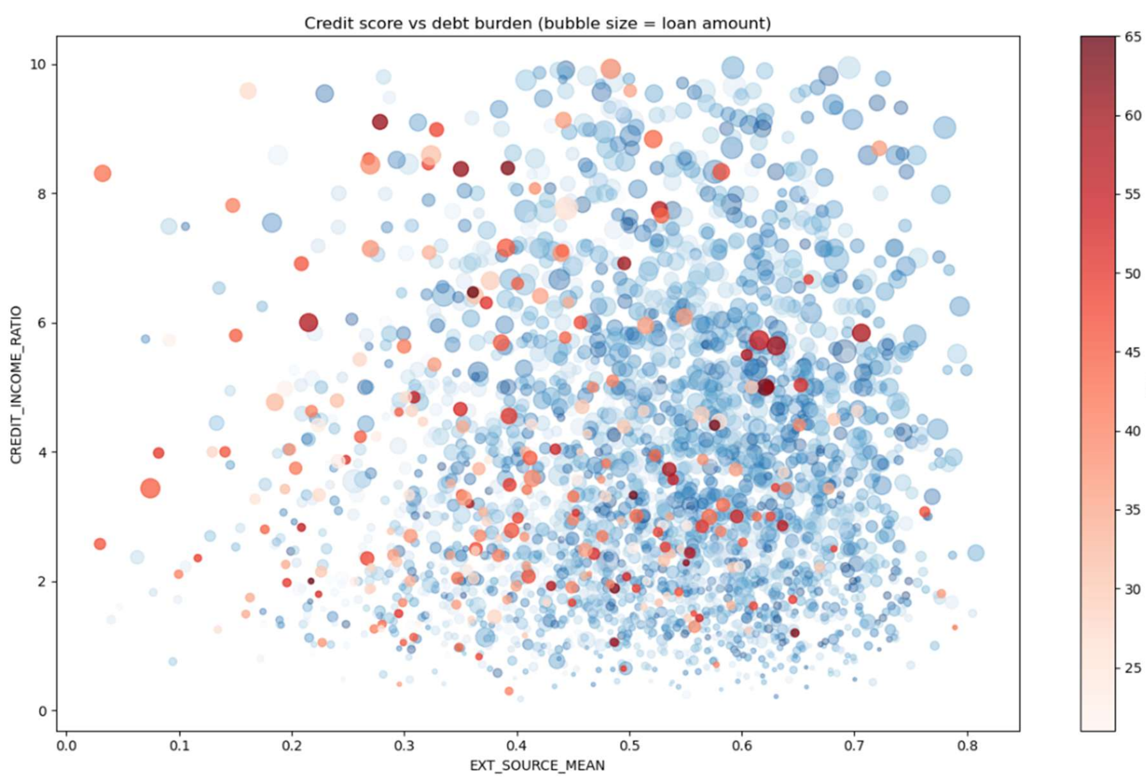
Introduction

This project focuses on predicting loan defaults for InclusiFinance. The goal is to identify borrower characteristics that signal higher default risk and recommend a model that balances predictive performance with interpretability. The broader objective is to help the company manage risk while still providing credit to customers who can benefit.

Data & Methodology

We used historical loan data including borrower demographics, credit history, and loan details. Because defaults are much less common than non-defaults, we applied oversampling to balance the dataset. The data was split 75/25 with stratification to preserve class balance. Models tested included Random Forest, XGBoost, a standard neural network (MLP), and a calibrated MLP. Class weights were applied in Random Forest and the scale_pos_weight parameter in XGBoost to address imbalance. Hyperparameters were tuned for each model, including learning rate, maximum depth, and number of trees for XGBoost, number of trees for Random Forest, and 2–3 hidden layers with 64–256 nodes plus early stopping for the MLPs. Model performance was evaluated using F1 score, precision, recall, AUC, a custom financial metric I created, training time, and interpretability through feature importance and SHAP values.

Exploratory analysis shows clear patterns in borrower risk. Defaults are concentrated where ext_source_mean is 0.3–0.5 and credit_income_ratio is 2–6, indicating a high-risk zone of moderate to low credit scores and higher debt. Defaults drop off above a credit score of 0.7, even with higher debt, though data is sparse in that region. Loan size, represented by bubble size, does not show a strong effect, suggesting it is not a primary driver of risk. Borrowers with higher credit scores rarely default, even if they carry more debt.



Results & Analysis
Model performance varied across the board. XGBoost slightly outperformed the others, with an AUC around 0.764, compared with Random Forest at 0.744 and the MLPs at 0.745. This means XGBoost ranks potential defaults more effectively and gives slightly clearer risk signals across probability thresholds. Random Forest trained faster than XGBoost and the MLPs, taking moderate time to fit. XGBoost required longer but remained practical, while the neural networks were more computationally intensive, especially with multiple hidden layers and early stopping. The calibrated MLP was slower to tune and harder to use operationally.

The threshold analysis shows the tradeoff between managing risk and maximizing financial impact. At an F1-optimal threshold of 0.65, XGBoost flags roughly 37 percent of defaults while approving 95 percent of applicants, generating a net value of about $153 million. At the financial-optimal threshold of 0.70, net value rises to roughly $202 million, catching around 30 percent of defaults and approving 93 percent of applicants. Very high thresholds near 0.85 flag few defaults and approve almost everyone, dropping net value to roughly $97 million. MLP Raw peaks around $171 million at 0.70 and retains flexibility across probability thresholds, while the calibrated MLP peaks at $171 million at a low threshold near 0.20 and flags almost no defaults above 0.50, making it less practical. Random Forest underperforms in both predictive accuracy and net financial impact. Feature importance from XGBoost highlights ext_source_mean and ext_source_by_age as the top drivers of default risk.

Model Comparison & Discussion
Choosing the right model depends on InclusiFinance's priorities. XGBoost offers the best balance between predictive performance, net value, and operational practicality. It captures non-linear relationships, ranks risk effectively, and can be tuned across thresholds to balance default coverage and approvals. Its ability to adjust the threshold allows the company to prioritize either approving more applicants or maximizing financial return, depending on their strategy. XGBoost also provides clear feature importance insights that can guide operational decisions, like monitoring high-risk borrower profiles or adjusting lending limits. MLP Raw is a reasonable backup, but requires more computational resources and is less interpretable. Calibrated MLP is difficult to use because its probability outputs are skewed, which makes high thresholds unreliable in practice. Random Forest is fast and interpretable but doesn't deliver anything special compared to the rest.

Conclusion & Future Steps
Deploying XGBoost around a threshold of 0.70 gives InclusiFinance the best balance between flagging potential defaults and approving qualified borrowers. This approach maximizes financial value while still capturing a meaningful portion of high-risk cases, helping the company manage risk without unnecessarily restricting access to credit. Moving forward, the model should be monitored and recalibrated as borrower behavior evolves, and additional data sources could be incorporated to improve predictions and ensure fairness. The threshold can be adjusted depending on whether the focus is on serving more applicants or prioritizing profit, giving flexibility in operational decision-making. The model's feature importance and threshold analysis can also guide operational policies, like targeted borrower outreach or credit limit adjustments, to further reduce risk and optimize returns.